ADVENTIST UNIVERSITY OF CENTRAL AFRICA

SOFTWARE TESTING ASSIGNMENT
Employee Management System
Black-Box Test Cases Report

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Course: STT Group-E

Item	Link / detail
GitHub repository	https://github.com/your-username/your-repo
Language	Java 17
Testing framework	JUnit 5
Test result	7 / 7 test cases passed

1. Introduction

This report documents the black-box testing of an Employee Management System built for Organization X. The system registers employees, calculates their payroll (including tax), and produces employee reports. Test cases were designed from the functional requirements only, without looking at the source code, using Equivalence Partitioning and Boundary Value Analysis.

2. Functional Requirements

The test cases in this report are traced to the following requirements:

ID	Functional requirement	Simple explanation
FR-1	Manage employee records	The system shall register a valid employee and retrieve it by ID. The employee name must not be empty and the base salary cannot be negative.
FR-2	Process payroll	The system shall compute gross pay (salary + bonus), apply a 10% tax below $30,000 and 20% from $30,000 upward, and return the net pay. Inactive employees cannot be paid.
FR-3	Generate report	The system shall produce a report of employees, and clearly state when no employee records exist.

3. Implementation & Test Environment

Language	Java 17
Testing framework	JUnit 5
Test type	Black-box (functional) testing
Techniques	Equivalence Partitioning, Boundary Value Analysis

4. Black-Box Testing (Functional) - Summary Table

Test Case ID	Req ID	Test Description & Scope	Technique	Input Specifications	Expected Output	Actual Output	Status
TC-BB-01	**FR-1**	Valid Employee Creation	Equivalence Partitioning (Valid)	ID='EMP001', Name='Alice Smith', Dept='Engineering', Salary=65000.0, Active=true	Employee record stored; retrievable by ID without errors.	Employee record stored & retrieved matching name 'Alice Smith' and $65,000.00 base salary.	**PASS**
TC-BB-02	**FR-1**	Invalid Base Salary Validation	Boundary Value Analysis (Boundary: -$500.00)	ID='EMP002', Name='Bob Jones', BaseSalary=-500.0	System throws IllegalArgumentException ('Base salary cannot be negative.')	IllegalArgumentException caught successfully. Record creation blocked.	**PASS**
TC-BB-03	**FR-1**	Empty Employee Name Validation	Equivalence Partitioning (Invalid)	ID='EMP003', Name=' ' (spaces), BaseSalary=50000.0	System throws IllegalArgumentException ('Employee name cannot be null or empty.')	IllegalArgumentException caught successfully. Record creation blocked.	**PASS**
TC-BB-04	**FR-2**	Valid Payroll Net Pay Calculation	Equivalence Partitioning (Standard Payroll)	Salary=50000.0, Bonus=10000.0 (Gross=60000.0 -> Tax Tier 20%)	Gross=$60,000.00, Tax=$12,000.00, Net Pay=$48,000.00	Payment generated with Gross=$60,000.00, Tax=$12,000.00, Net Pay=$48,000.00.	**PASS**
TC-BB-05	**FR-2**	Tax Bracket Threshold Boundary	Boundary Value Analysis ($30,000 Boundary)	Salary 1 = $29,999.99; Salary 2 = $30,000.00	Salary 1 yields 10% tax rate; Salary 2 yields 20% tax rate.	Rate for $29,999.99 = 0.10 (10%); Rate for $30,000.00 = 0.20 (20%). Boundary verified.	**PASS**
TC-BB-06	**FR-2**	Payment for Inactive Employee	Equivalence Partitioning (Inactive State)	Employee ID='EMP006', Active=false, Bonus=0.0	System throws IllegalStateException ('Cannot process payment for inactive employee')	IllegalStateException caught cleanly. Payment disbursement blocked.	**PASS**
TC-BB-07	**FR-3**	Report Generation on Empty Database	Equivalence Partitioning (Empty Dataset)	Employee Database with 0 records.	Report generated containing 'No employee records found in system.'	Report generated clean with zero errors, outputting graceful empty status message.	**PASS**

5. Detailed Black-Box Test Case Reports

Each report follows the structure of the test-case-report template.

TEST CASE REPORT: TC-BB-01 - Valid Employee Creation

GENERAL INFORMATION	
Test Stage	[X] Functionality [ ] Unit [ ] Integration
Test Date / System Date	09/22/2026
Tester Name	Solide AZE
Test Case Number	**TC-BB-01**
Test Case Description	Valid Employee Creation
Test Technique	Equivalence Partitioning (Valid)
Result Status	**[X] Pass [ ] Fail**
Incident Number	N/A
INTRODUCTION & REQUIREMENTS	
Requirement(s) Tested	FR-1: Functional Specification Requirement
Roles & Responsibilities	Test Engineer: Execute functional cases. Developer: Fix identified defects.
Setup & Stop Procedures	Setup: Initialize JDK 17 & Service Repositories. Stop: Tear down test context.
TEST SPECIFICATIONS & RESULTS	
Input Specifications	ID='EMP001', Name='Alice Smith', Dept='Engineering', Salary=65000.0, Active=true
Expected Results	Employee record stored; retrievable by ID without errors.
Actual Results	Employee record stored & retrieved matching name 'Alice Smith' and $65,000.00 base salary.
Final Disposition	**PASS - All functional assertion criteria satisfied with 0 defects.**

TEST CASE REPORT: TC-BB-02 - Invalid Base Salary Validation

GENERAL INFORMATION	
Test Stage	[X] Functionality [ ] Unit [ ] Integration
Test Date / System Date	09/22/2026
Tester Name	Solide AZE
Test Case Number	**TC-BB-02**
Test Case Description	Invalid Base Salary Validation
Test Technique	Boundary Value Analysis (Boundary: -$500.00)
Result Status	**[X] Pass [ ] Fail**
Incident Number	N/A
INTRODUCTION & REQUIREMENTS	
Requirement(s) Tested	FR-1: Functional Specification Requirement
Roles & Responsibilities	Test Engineer: Execute functional cases. Developer: Fix identified defects.
Setup & Stop Procedures	Setup: Initialize JDK 17 & Service Repositories. Stop: Tear down test context.
TEST SPECIFICATIONS & RESULTS	
Input Specifications	ID='EMP002', Name='Bob Jones', BaseSalary=-500.0
Expected Results	System throws IllegalArgumentException ('Base salary cannot be negative.')
Actual Results	IllegalArgumentException caught successfully. Record creation blocked.
Final Disposition	**PASS - All functional assertion criteria satisfied with 0 defects.**

TEST CASE REPORT: TC-BB-03 - Empty Employee Name Validation

GENERAL INFORMATION	
Test Stage	[X] Functionality [ ] Unit [ ] Integration
Test Date / System Date	09/22/2026
Tester Name	Solide AZE
Test Case Number	**TC-BB-03**
Test Case Description	Empty Employee Name Validation
Test Technique	Equivalence Partitioning (Invalid)
Result Status	**[X] Pass [ ] Fail**
Incident Number	N/A
INTRODUCTION & REQUIREMENTS	
Requirement(s) Tested	FR-1: Functional Specification Requirement
Roles & Responsibilities	Test Engineer: Execute functional cases. Developer: Fix identified defects.
Setup & Stop Procedures	Setup: Initialize JDK 17 & Service Repositories. Stop: Tear down test context.
TEST SPECIFICATIONS & RESULTS	
Input Specifications	ID='EMP003', Name=' ' (spaces), BaseSalary=50000.0
Expected Results	System throws IllegalArgumentException ('Employee name cannot be null or empty.')
Actual Results	IllegalArgumentException caught successfully. Record creation blocked.
Final Disposition	**PASS - All functional assertion criteria satisfied with 0 defects.**

TEST CASE REPORT: TC-BB-04 - Valid Payroll Net Pay Calculation

GENERAL INFORMATION	
Test Stage	[X] Functionality [ ] Unit [ ] Integration
Test Date / System Date	09/22/2026
Tester Name	Solide AZE
Test Case Number	**TC-BB-04**
Test Case Description	Valid Payroll Net Pay Calculation
Test Technique	Equivalence Partitioning (Standard Payroll)
Result Status	**[X] Pass [ ] Fail**
Incident Number	N/A
INTRODUCTION & REQUIREMENTS	
Requirement(s) Tested	FR-2: Functional Specification Requirement
Roles & Responsibilities	Test Engineer: Execute functional cases. Developer: Fix identified defects.
Setup & Stop Procedures	Setup: Initialize JDK 17 & Service Repositories. Stop: Tear down test context.
TEST SPECIFICATIONS & RESULTS	
Input Specifications	Salary=50000.0, Bonus=10000.0 (Gross=60000.0 -> Tax Tier 20%)
Expected Results	Gross=$60,000.00, Tax=$12,000.00, Net Pay=$48,000.00
Actual Results	Payment generated with Gross=$60,000.00, Tax=$12,000.00, Net Pay=$48,000.00.
Final Disposition	**PASS - All functional assertion criteria satisfied with 0 defects.**

TEST CASE REPORT: TC-BB-05 - Tax Bracket Threshold Boundary

GENERAL INFORMATION	
Test Stage	[X] Functionality [ ] Unit [ ] Integration
Test Date / System Date	09/22/2026
Tester Name	Solide AZE
Test Case Number	**TC-BB-05**
Test Case Description	Tax Bracket Threshold Boundary
Test Technique	Boundary Value Analysis ($30,000 Boundary)
Result Status	**[X] Pass [ ] Fail**
Incident Number	N/A
INTRODUCTION & REQUIREMENTS	
Requirement(s) Tested	FR-2: Functional Specification Requirement
Roles & Responsibilities	Test Engineer: Execute functional cases. Developer: Fix identified defects.
Setup & Stop Procedures	Setup: Initialize JDK 17 & Service Repositories. Stop: Tear down test context.
TEST SPECIFICATIONS & RESULTS	
Input Specifications	Salary 1 = $29,999.99; Salary 2 = $30,000.00
Expected Results	Salary 1 yields 10% tax rate; Salary 2 yields 20% tax rate.
Actual Results	Rate for $29,999.99 = 0.10 (10%); Rate for $30,000.00 = 0.20 (20%). Boundary verified.
Final Disposition	**PASS - All functional assertion criteria satisfied with 0 defects.**

TEST CASE REPORT: TC-BB-06 - Payment for Inactive Employee

GENERAL INFORMATION	
Test Stage	[X] Functionality [ ] Unit [ ] Integration
Test Date / System Date	09/22/2026
Tester Name	Solide AZE
Test Case Number	**TC-BB-06**
Test Case Description	Payment for Inactive Employee
Test Technique	Equivalence Partitioning (Inactive State)
Result Status	**[X] Pass [ ] Fail**
Incident Number	N/A
INTRODUCTION & REQUIREMENTS	
Requirement(s) Tested	FR-2: Functional Specification Requirement
Roles & Responsibilities	Test Engineer: Execute functional cases. Developer: Fix identified defects.
Setup & Stop Procedures	Setup: Initialize JDK 17 & Service Repositories. Stop: Tear down test context.
TEST SPECIFICATIONS & RESULTS	
Input Specifications	Employee ID='EMP006', Active=false, Bonus=0.0
Expected Results	System throws IllegalStateException ('Cannot process payment for inactive employee')
Actual Results	IllegalStateException caught cleanly. Payment disbursement blocked.
Final Disposition	**PASS - All functional assertion criteria satisfied with 0 defects.**

TEST CASE REPORT: TC-BB-07 - Report Generation on Empty Database

GENERAL INFORMATION	
Test Stage	[X] Functionality [ ] Unit [ ] Integration
Test Date / System Date	09/22/2026
Tester Name	Solide AZE
Test Case Number	**TC-BB-07**
Test Case Description	Report Generation on Empty Database
Test Technique	Equivalence Partitioning (Empty Dataset)
Result Status	**[X] Pass [ ] Fail**
Incident Number	N/A
INTRODUCTION & REQUIREMENTS	
Requirement(s) Tested	FR-3: Functional Specification Requirement
Roles & Responsibilities	Test Engineer: Execute functional cases. Developer: Fix identified defects.
Setup & Stop Procedures	Setup: Initialize JDK 17 & Service Repositories. Stop: Tear down test context.
TEST SPECIFICATIONS & RESULTS	
Input Specifications	Employee Database with 0 records.
Expected Results	Report generated containing 'No employee records found in system.'
Actual Results	Report generated clean with zero errors, outputting graceful empty status message.
Final Disposition	**PASS - All functional assertion criteria satisfied with 0 defects.**